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Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
 - Profile setup: Create a user account (e.g., server / server123)
 - Install OpenSSH server: **Check this option** (Important!)
 - Selected software: **None** (Standard Ubuntu Server)
3. Wait for installation to complete and reboot

Network Configuration

Configure Network Mode

For VirtualBox:

1. Power off the VM (if running)
2. Go to **Settings** → **Network**
3. **Adapter 1**: Enable Network Adapter
4. **Attached to**: Select **Bridged Adapter**
5. **Name**: Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM
2. Go to **VM Settings** → **Network Adapter**

3. Select **Bridged** mode

4. Click **OK**

Install/Verify SSH Server

Server: Install SSH on the server (if not already installed):

- `sudo systemctl status ssh` – for checking the SSH status.
- `sudo apt update` – updating the your server.
- `sudo apt install openssh-server -y` –installing the SSH.
- `sudo systemctl enable ssh` – enabling the SSH to your unit. (SERVER/DESKTOP)
- `sudo systemctl start ssh` – starting the SSH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

Inside the VM, run:

```
Sample@UBUNTU-DESKTOP:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9b:8c:ca brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.4/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83602sec preferred_lft 83602sec
    inet6 2001:4451:b00:2000:d76c:1af2:a10c:f36d/64 scope global temporary dynamic
        valid_lft 179068sec preferred_lft 83386sec
    inet6 2001:4451:b00:2000:a00:27ff:fe9b:8cca/64 scope global dynamic mngtmpadr proto kernel_r
        valid_lft 179068sec preferred_lft 92668sec
    inet6 fe80::a00:27ff:fe9b:8cca/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
Sample@UBUNTU-DESKTOP:~$
```

From the desktop, connect to the server: `ssh username@192.168.1.120`

2. Connect from Desktop

From your host desktop:

```
Sample@UBUNTU-DESKTOP:~$ ssh Server@192.168.1.14
Server@192.168.1.14's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep  8 11:24:38 PM UTC 2025

System load:          0.0
Usage of /:           5.1% of 48.91GB
Memory usage:         5%
Swap usage:           0%
Processes:            139
Users logged in:      1
IPv4 address for enp0s3: 192.168.1.14
IPv6 address for enp0s3: 2001:4451:b00:2000:a00:27ff:fe57:bc4a

Expanded Security Maintenance for Applications is not enabled.

136 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep  8 22:52:07 2025 from 192.168.1.4
Server@UBUNTU-SERVER:~$
```

Submission Requirements

1. Screenshots (labeled clearly):

- VM network settings showing Bridged Adapter
- Server IP address (output of `ip a` or `hostname -I`)
- SSH service status (`sudo systemctl status ssh`)
- Successful SSH connection from host
- Verification commands executed inside VM

2. Submission Format:

- Single PDF file
- Include name, student ID, and date
- All screenshots with brief explanation
- Upload to your portfolio website.

UBUNTU SERVER SSH REMOTE ACCESS LAB

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Course/Year/Section: BSIT3B

Date Submitted: 02-09-26

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1 OBJECTIVES (5 POINTS)

Check all that apply:

- ☐ **Create and configure a Virtual Machine with Ubuntu Server**
- ☐ **Configure Bridged Network Adapter for network access**
- ☐ **Determine and verify the server's IP address**
- ☐ **Install OpenSSH Server and enable the service**
- ☐ **Establish successful SSH connection from host machine**
- ☐ **Execute verification commands to confirm remote access**

2 HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	Oracle Linux
Virtualization Software (VirtualBox/VMware)	Vm Virtual Bos
Ubuntu Server Version	Ubuntu 26.04 LTS
VM Name	BubanElectiveServer
RAM Allocated	4
Storage Allocated	10048
Network Mode	Host Only Adapter
SSH Client Used	Secure shell client

3 VM CONFIGURATION

Setting	Value
VM Name	UBUNTU-SERVER
OS Type	Ubuntu (64-bit)
Memory	10048 MB
Hard Disk	108 GB
Network Adapter	Bridged

4. NETWORK CONFIGURATION

6. REMOTE CONNECTION Connection Details:

The screenshot shows a Kali Linux desktop environment. The main window is the Oracle VM VirtualBox interface, displaying the 'BubunEctive3 - Settings' dialog box. The 'Network' tab is selected, showing the configuration for 'Adapter 1'. The 'Enable Network Adapter' checkbox is checked, and the 'Attached to' dropdown is set to 'NAT'. The 'Name' field is set to 'enp0s3'. The 'Advanced' section is expanded. The background shows the Kali Linux desktop with various application icons on the left and a terminal window at the bottom displaying network configuration commands and output.

```
File Machine View Input Devices Help
reated symlink '/etc/systemd/system/sshd.service' -> '/usr/lib/systemd/system/ssh.service'.
reated symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' -> '/usr/lib/systemd/system/ssh.service'.
anebubandane:"$ sudo systemctl start ssh
udp> "systemctl start ssh": command not found
anebubandane:"$ sudo systemctl enable ssh
ppport-coredump-hook@
ppport-forward@
utov@
lockdev@
ssd@
hrony-dnssrv@
hrony-dnssrv.timer
hrony-wait.service
ontainer-getty.service
sh-session1.service
2scrub@
2scrub-fail@
xit.target
etty@
ait.target
ssid.service
exec.target
xd-installer@
admi-grow-continue@
admi-last-resort@
dmone
udmabee
ftables.service
oikit-agent-helper@
oweroff.target
roc-sys-fs-binfmt_misc.mount
votaon@
anebubandane:"$ ip net
bject "ip" is unknown, try "ip help".
anebubandane:"$ ip a
: lo:<LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/backbone 00:00:00:00:00:00 bd 00:00:00:00:00:00
inet 127.0.0.1/b scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
: enpos3:<BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
link/ether 00:00:27:96:30:a0 bd ff:ff:ff:ff:ff:ff
altname enx0000279630a0
inet 192.168.56.101/24 metric 100 bd 192.168.56.255 scope global dynamic enpos3
valid_lft 593sec preferred_lft 593sec
inet6 fe80::a00:27fff:fef3:30a0/64 scope link proto kernellll
valid_lft forever preferred_lft forever
anebubandane:"$
```

After i login and put command ip a so that i can see my ip address

```
inet6 fe80::a00:27ff:fe98:300b/64 scope link proto kernel llmnr
valid_lft forever preferred_lft forever
anebuban@dane:~$ sudo systemctl status ssh
sudo: authenticate] Password:
ssh.service - OpenBSD Secure Shell server
Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
Active: active (running) since Wed 2026-09-02 06:34:21 UTC; 1h 52min ago
Invocation: 3946fab5cabe4dcea8677566973e1f9c
TriggeredBy: ● ssh.socket
Docs: man:sshd(8)
      man:sshd_config(5)
Main PID: 1374 (sshd)
Tasks: 1 (limit: 8196)
Memory: 5.2M (peak: 7.5M)
CPU: 211ms
CGroup: /system.slice/ssh.service
        └─1374 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
```

Then i put a command ssh so if its working



```
bunantictiveServer [running] - Oracle VM VirtualBox
File Machine View Input Devices Help
buntu 26.04 LTS dane tty1
ane login: danebuban
assword:
elcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Wed Sep  2 08:28:04 AM UTC 2026

System load: 0.0          Memory usage: 3%   Processes:    151
Usage of /:  13.8% of 52.07GB Swap usage:   0%   Users logged in: 0

xpanded Security Maintenance for Applications is not enabled.

3 updates can be applied immediately.
o see these additional updates run: apt list --upgradable

nable ESM Apps to receive additional future security updates.
ee https://ubuntu.com/esm or run: sudo pro status

anebuban@dane:~$
```

Welcome

```
danebuban@dane: ~ — ssh danebuban@192.168.56.1...
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Sep  2 07:05:22 2026 from 192.168.56.1
danebuban@dane:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP
   group default qlen 1000
   link/ether 08:00:27:96:30:db brd ff:ff:ff:ff:ff:ff
   altname enx0800279630db
   inet 192.168.56.101/24 metric 100 brd 192.168.56.255 scope global dynamic en
   p0s3
       valid_lft 459sec preferred_lft 459sec
   inet6 fe80::a00:27ff:fe96:30db/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
danebuban@dane:~$ =
```

The i put a command to my terminal in desktop ubuntu to connect it.